

NITEKELA SECONDARY SCHOOL

FORM TWO SATURDAY EXAMINATION SERIES #2

CHEMISTRY MARKING GUIDE

SECTION A

1. @ 01 = 10 Marks

I	ii	iii	iv	v	vi	vii	viii	ix	x
C	B	C	C	B	C	B	B	B	B

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2. @ 01 = 05 Marks

I	ii	iii	iv	v
C	A	F	G	E

SECTION B

3. (a) (i) 01 Mark

LATIN NAMES or GREEK NAMES

(ii) 03 Marks (any three)

ENGLISH NAME	LATIN NAME
Antimony	Stibium
Copper	Cuprum
Gold	Aurum
Iron	Ferrum
Lead	Plumbum
Mercury	Hydrargyrum
Potassium	Kalium
Sodium	Natrium
Silver	Argentum
Tin	Stannum
Tungsten	Wolfram

(b) @ 01 = 03 Marks (three points)

- ❖ Quickly understand elements being referred to, instead of memorizing the full names.
- ❖ Help the work of writing chemical equations in abbreviated forms instead of writing each element in its full name.

❖ Distinguish one element from other.

(c) @ 01 = 03 Marks

(i) To prevent contamination of chemicals.

(ii) To allow proper ventilation.

(iii) For emergence purpose (Easy escaping during accident Example fire)

4. (a) (i) 03 Marks

Q = Li

L = Cl

(ii) 02 Marks

M = 2:8:2

C = 2:8

(iii) 01 Mark

D

(iv) 01 Mark

Q and H

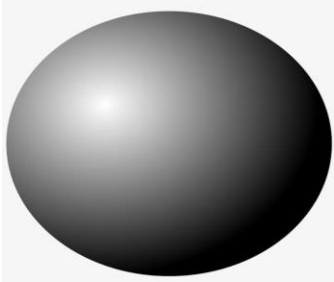
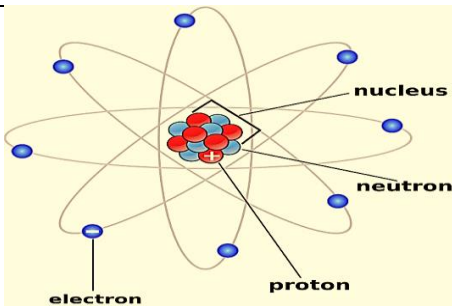
(v) 01 Mark (any one)

- ❖ Q, D and C
- ❖ H, M and L
- ❖ Q and C
- ❖ D and C
- ❖ Q and D
- ❖ H and M
- ❖ M and L
- ❖ H and L

(b) 02 Marks

Mendeleev's periodic law arranges elements in periodic table in order of increasing atomic mass While **Modern periodic law** arranges elements in periodic table in order of increasing atomic numbers

5. (a) 02 @ = 04 Marks

Dalton's Atomic Theory	Rutherford's Atomic modal
	

(b) @ 01 = 04 Marks (four points)

- ❖ Atoms can neither be created nor destroyed.
- ❖ Matter is made up of tiny particles called atoms
- ❖ Atoms of different element may combine in simple ratio.
- ❖ Atoms of the same element are identical and have same atomic masses.
- ❖ Atoms of different element are not identical and have different atomic masses.

(c) @ 01 = 02 Marks

- ❖ NEUTRONS
- ❖ PROTONS

6. (a) @ 01 = 06 Marks

$^{27}_{13}\text{Al}$	$^{235}_{92}\text{U}$
Protons = 13	Protons = 92
Neutrons = 14	Neutrons = 143
Electrons = 13	Electrons = 92

(b) @ 02 = 04 Marks (two points)

- ❖ Dating fossils
- ❖ Cancer treatment
- ❖ Medical diagnosis

7. (a) (i) 02 Marks

- ❖ FIRE CLASS A since burning materials were books and magazines (burning materials of class A)

(ii) 01 Mark

- ❖ Any portable fire extinguisher except CARBON DIOXIDE EXTINGUISHER

(iii) 02 Marks

- ❖ Carbon dioxide extinguisher Because fire can re-ignite since it removes oxygen temporarily. Also it has poor cooling effect

(iv) 02 Marks

- ❖ Putting off main switch

(b) @ 01 = 03 Marks (three points)

- ❖ They are very hard (can't break easily)
- ❖ They are very resistant to heat (can heat at high temperature)
- ❖ Not corroded easily
- ❖ Easy to wash (comperatively)
- ❖ Resistant to oxidation.

8. @ 01 = 05 Marks (five points)

- ❖ Manufacture of medicines.
- ❖ Manufacture of plastics.

- ❖ Manufacture of clothes and shoes.
- ❖ Manufacture of pesticides, insecticides and fungicides.
- ❖ Manufacture of cooking oil and sugar.
- ❖ Manufacture of construction materials like iron sheet and cement.
- ❖ Help to answer fundamental questions like why iron rust.
- ❖ Help to know chemical reactions taking place in our body.
- ❖ Manufacture of fuel like petrol, diesel and kerosine.
- ❖ Contribute to manufacture of electronic products like computers and smart phones. (Modern smart phones made of at least 15 elements including indium and iron)
- ❖ Help to get carrier and professions like environmentalist, doctor, nurse and pharmacist.

(b) @ 01 = 05 Marks

- ❖ Making thorough observations.
- ❖ Recoding accurately what has been observed.
- ❖ Organizing the observed and recorded information.
- ❖ Repeating tests to ensure observations.
- ❖ Drawing conclusions from observations.
- ❖ Predicting possible outcomes of similar experiments.

9. (a) @ 02 = 04 Marks

- ❖ Open windows and doors to ensure proper ventilation.
- ❖ Evacuate the area

(b) @ 0.5 = 03 Marks (six points)

- ❖ Doctor
- ❖ Nurse
- ❖ Teacher
- ❖ Pharmacist
- ❖ Chemical engineer
- ❖ Environmentalist
- ❖ Chemical analyst
- ❖ Laboratory technician
- ❖ Laboratory scientist

(c) @ 01 = 03 Marks (three points)

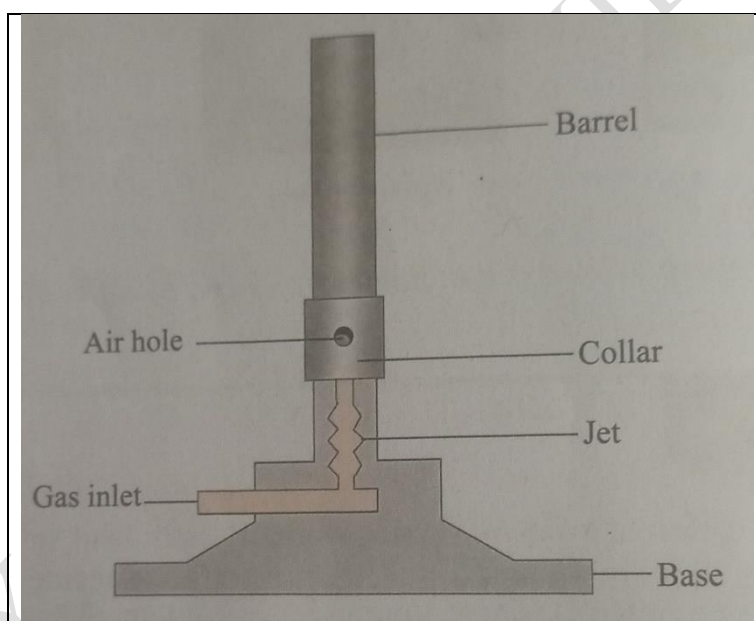
- ❖ Should have fume chamber
- ❖ Should have large windows and doors for proper ventilation
- ❖ Doors and windows should open outward
- ❖ Its floor should rough
- ❖ Should have preparation room
- ❖ Should have enough fire extinguishers
- ❖ Should have sewage system
- ❖ Should have water supply system
- ❖ Should have gas supply system
- ❖ Should have electric supply system.
- ❖ Should have enough space.
- ❖ Should have enough heat sources and other apparatus

SECTION C

10. (a) @ 1.5 = 06 Marks (four points)

Air hole closed (LUMINOUS FLAME)	Air holes open (NON-LUMINOUS FLAME)
Bright yellow	Blue
Produce soot and smoke	Do not produce soot and smoke
Produce less heat	Produce more heat
Has four zones	Has three zones
Not steady	Steady
Silent	Noisy

(b) 04 Marks (01 for heading, 01 for labeling and 02 for neat diagram)



(c) 05 Marks

- ❖ Connect the Bunsen burner by a rubber tube to the gas supply.
- ❖ Close the air holes.
- ❖ Turn the gas on to let in sufficient gas.
- ❖ Quickly bring flame at the top of the barrel.
- ❖ Turn the collar to adjust air holes until you get type of flame you want.

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“Enjoy your life, learn to forgive and forget, hatred may reduce your life span”

Sir Medrack Mponzi